

ELEC5305 Project Proposal Guide

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1 Project Title

Speaker Anonymisation Using Classical Signal Processing and Zero-Shot Neural Voice Conversion

2 Student Information

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3 Project Overview

Speech recordings contain not only linguistic information but also characteristics that can reveal the identity of the speaker. This creates privacy concerns when speech is stored, transmitted, or used by speech-based AI systems. Voice anonymisation aims to modify speaker-specific characteristics while preserving the linguistic content and intelligibility of the speech.

This project will investigate and compare two approaches to speaker anonymisation: a classical signal-processing method based on linear predictive coding (LPC) and spectral-envelope modification, and a modern zero-shot neural voice conversion method based on Seed-VC.

The classical system will be implemented in MATLAB using concepts related to the source-filter model and LPC covered in ELEC5305, while the neural system and evaluation pipeline will be implemented in Python using PyTorch.

4 Background and Motivation

A speech signal contains information about both the spoken message and the speaker. Voice anonymisation attempts to modify the speaker-dependent characteristics of speech so that automatic speaker recognition becomes less reliable while useful information such as linguistic content remains available.

The VoicePrivacy research initiative has established privacy–utility evaluation as an important framework for speaker anonymisation. Existing approaches include signal-processing transformations, speaker-embedding manipulation, neural speech synthesis, and voice conversion. Classical approaches based on LPC and spectral-envelope modification have the advantage of low computational cost and interpretability.

More recent approaches use learned speech representations and neural waveform generation to replace speaker identity while retaining speech content. Seed-VC is a recent zero-shot voice conversion framework based on diffusion transformers that separates source speech content from target speaker timbre and can perform conversion without retraining for each speaker.

This topic was selected because I have some experience of deep learning network and I am more familiar with Python. So I would like to try neural network in acoustic field. It connects classical speech signal processing with modern deep learning.

5 Proposed Methodology

The project will use both MATLAB and Python.

The classical anonymization baseline will be developed in MATLAB. Speech will be segmented into short frames and analyzed using LPC. The LPC poles or spectral envelope will then be modified using a McAdams-style transformation before speech resynthesis. Different transformation strengths will be tested to investigate the relationship between voice modification and intelligibility.

The neural anonymization system will be implemented in Python using a pretrained Seed-VC model. Rather than training a large model from scratch, the project will investigate several inference strategies, including the model's anonymization mode and conversion using different target speakers. If time permits, target speakers with different acoustic characteristics will be selected to examine whether target selection affects anonymization performance.

A common evaluation pipeline will be developed in Python. Privacy will be assessed using a pretrained speaker-verification model such as ECAPA-TDNN by comparing speaker embeddings or verification performance before and after anonymization. ECAPA-TDNN is a well-established neural architecture for speaker verification.

Speech utility will primarily be evaluated using automatic speech recognition and word error rate (WER). Additional signal-level analysis may include spectrogram comparison, fundamental-frequency analysis, and processing time.

A small public speech dataset such as LibriSpeech or VCTK will be used. Only a manageable subset of speakers and utterances will be required so that the experiments remain feasible within the semester.

6 Expected Outcomes

The expected outcome is a working speaker-anonymization evaluation system containing both a classical MATLAB implementation and a deep-learning Python implementation.

The project is expected to produce quantitative comparisons of privacy and intelligibility for the different anonymization methods. Speaker-verification performance or speaker similarity will be used as a privacy measure, while WER will measure preservation of linguistic information. Audio examples and spectrograms will also be provided so that differences can be demonstrated.

The final GitHub repository will contain source code, experiment scripts, documentation, selected anonymized audio examples, figures, the written report, and a GitHub Pages project website.

7 Timeline (Weeks 1–13)

Week	Task
1–2	Topic selection, GitHub setup, initial literature review
3–5	Dataset preparation and environment setup
6–9	Implementation of classical and neural anonymisation methods
10–11	Evaluation, comparison, and analysis
12–13	Final report, GitHub documentation

8 References

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